Quantification: I534_z25c1-2
============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	14
interior lacuna count	13
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 13)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	2941.54	2951	1607.61	px^2
major_axis_length_px	lacuna shape	85.8598	83.1379	30.1528	px
minor_axis_length_px	lacuna shape	45.6621	45.1555	14.9313	px
aspect_ratio	lacuna shape	1.9335	1.9125	0.5477	unitless
eccentricity	lacuna shape	0.8143	0.8524	0.1082	unitless
solidity	lacuna shape	0.8338	0.8509	0.098	unitless
roots_count	headline	7.3846	6	4.8741	count
ring_length_r30_px	headline	255.385	231	125.893	px
ring_length_r60_px	reported	592.462	539	189.253	px
owned_length_px	ownership-dependent	736.109	586.316	577.88	px
edge_count	ownership-dependent	34.6154	24	26.5943	count
mean_edge_length_px	ownership-dependent	21.2869	20.1007	4.3991	px
perimeter_px	normalised	273.55	262.877	100.444	px
ring_area_r30_px2	normalised	9892.77	9533	2491.02	px^2
ring_area_r60_px2	normalised	23185.9	22663	4361.73	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.99	1	0.0267	unitless
ring_density_r30	normalised	0.0250278	0.0231267	0.00825029	px^-1
ring_density_r60	normalised	0.0255117	0.0236435	0.00652795	px^-1
roots_per_100px_perimeter	normalised	2.508	2.5172	0.88	per 100 px
ring_attached_length_r30_px	network v2	218.077	198	127.263	px
ring_attached_length_r60_px	network v2	400.231	373	214.829	px
ring_length_w_r30_px	network v2	290.582	258.309	144.07	px
ring_length_w_r60_px	network v2	675.121	606.973	213.742	px
sholl_crossings_r10	network v2	6	6	2.6458	count
sholl_crossings_r20	network v2	7.3846	6	4.0525	count
sholl_crossings_r30	network v2	8.3077	8	3.4733	count
loop_count	loops, report only	1.7692	1	2.4884	count
loop_edge_share	loops, report only	0.1322	0.0678	0.1566	share
ring_width_mean_r30_px	width	7.7515	7.6554	0.5396	px

Per field
----

measure	value	units
analysed_area_px2	1.00762e+06	px^2
total_skeleton_length_px	23698	px
canalicular_length_density_per_px	0.0235189	px^-1
junction_count	439	count
junction_density_per_px2	0.00043568	px^-2
lacuna_count	14	count

lacunae_per_px2	1.389e-05	px^-2
skeleton_component_count	348	count
mean_component_length_px	68.0977	px
median_component_length_px	21	px
field_length_density_w_per_px	0.0266949	px^-1
field_density_without_flagged_per_px	0.0255902	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	245	px
unattached_component_count	16	count
canalicular_length_density_attached_per_px	0.0235189	px^-1
field_length_density_with_unattached_per_px	0.023762	px^-1
loop_count	36	count
loop_edge_share	0.0981	share
edge_count	1213	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.8493	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	22343	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	718	47.2395	23.002	2.0537	0.8734	0.8022	-1.0816	93.7911	437.035	False
2	2720	92.088	38.5852	2.3866	0.908	0.9547	-1.1428	215.941	986.257	True
3	6078	103.767	76.7807	1.3515	0.6727	0.9519	-0.9596	258.575	697.738	False
4	2951	92.7411	57.1766	1.622	0.7873	0.6519	0.6626	412.099	797.415	False
5	3683	112.658	46.4293	2.4264	0.9111	0.7914	-0.9592	433.344	623.77	False
6	3483	74.1218	60.6446	1.2222	0.575	0.9286	-0.204	483.984	66.1174	False
7	3784	85.9898	57.0703	1.5067	0.748	0.9309	-0.1766	493.888	417.965	False
8	4829	127.96	50.7336	2.5222	0.918	0.7542	-1.2747	583.146	228.783	False
9	2220	83.1379	43.4719	1.9125	0.8524	0.7754	0.0297	577.841	853.017	False
10	4318	144.296	45.1555	3.1955	0.9498	0.6869	1.4075	580.847	622.689	False
11	2037	75.2328	35.6451	2.1106	0.8806	0.9122	-0.2022	721.922	650.957	False
12	1103	54.1002	27.7589	1.9489	0.8583	0.9026	-0.7329	709.348	70.3019	False
13	1992	69.0062	38.1155	1.8104	0.8336	0.9009	-1.2002	817.186	92.8996	False
14	1044	45.928	31.6235	1.4523	0.7252	0.8509	0.2821	817.663	289.875	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	2	231	643	1631.07	87	18.7479	8.2543
2	3	55	210	61.7696	5	12.3539	6.4488
3	6	275	685	261.309	13	20.1007	7.3739
4	13	460	1038	1853.7	81	22.8852	7.061
5	11	312	500	304.836	16	19.0522	7.5715

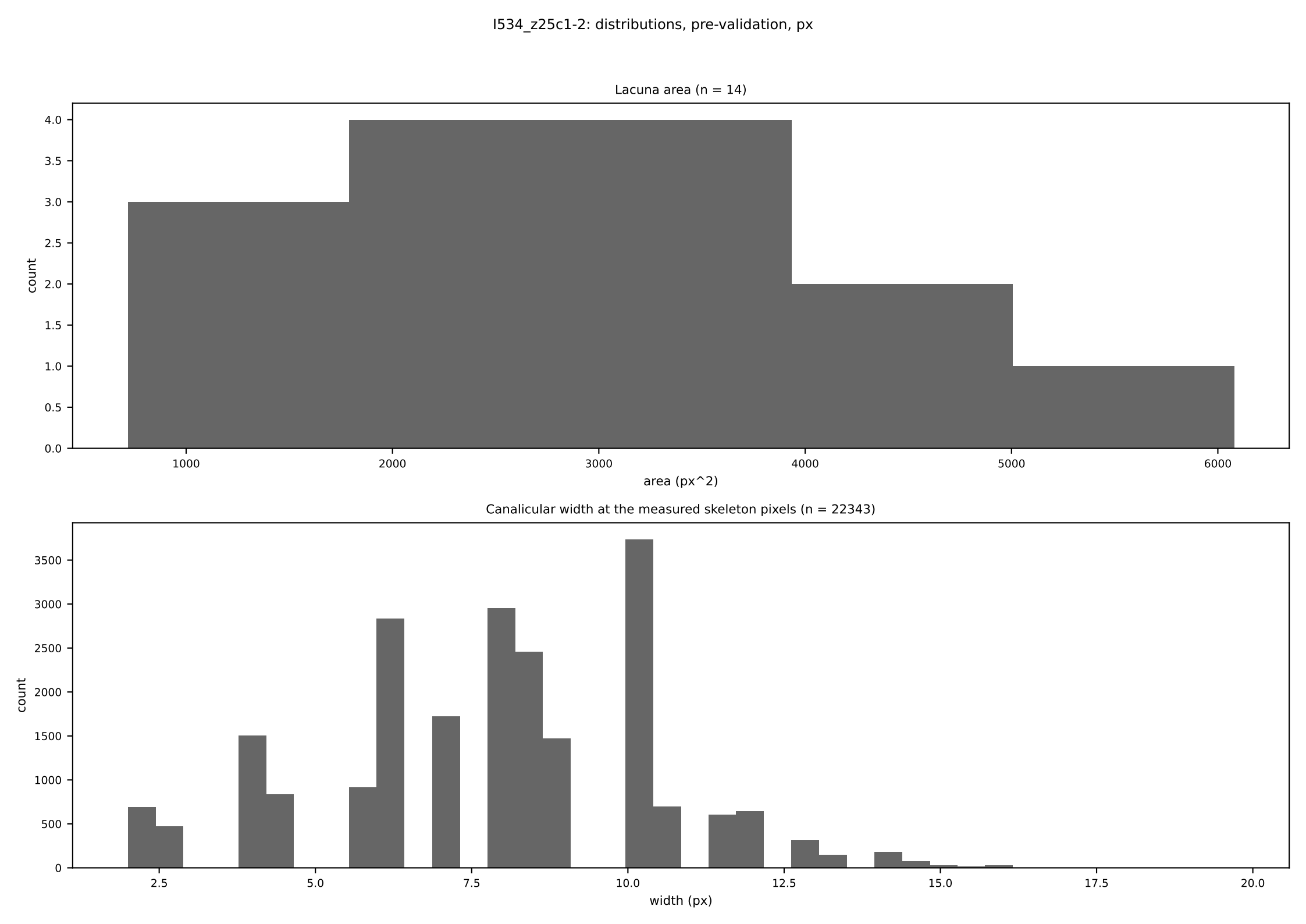
6	6	175	539	354.137	21	16.8637	8.5517
7	4	157	533	184.681	12	15.3901	7.7554
8	18	496	814	820.411	50	16.4082	7.4463
9	7	270	645	842.909	34	24.7914	7.6554
10	13	376	691	683.6	31	22.0516	8.0545
11	3	147	421	1516.09	59	25.6965	8.8704
12	4	125	390	252.037	13	19.3874	7.2836
13	6	190	388	586.316	24	24.4298	7.2251
14	3	106	415	278.321	9	30.9245	7.6667

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	132.125	6322	18207	1	1	0.0365391	0.0353161	1.5137
2	237.965	7174	17084	0.7631	0.6986	0.00766657	0.0122922	1.2607
3	334.475	11891	27068	1	1	0.0231267	0.0253066	1.7939
4	419.345	11919	24709	1	1	0.0385938	0.042009	3.1001
5	332.676	11820	24257	1	1	0.0263959	0.0206126	3.3065
6	249.078	9533	22009	1	0.9067	0.0183573	0.02449	2.4089
7	262.877	9914	25300	1	1	0.0158362	0.0210672	1.5216
8	405.588	13364	31075	1	1	0.0371146	0.0261947	4.438
9	278.09	9508	22663	1	1	0.0283971	0.0284605	2.5172
10	406.416	13768	29555	1	1	0.0273097	0.0233801	3.1987
11	220.35	8451	20862	1	1	0.0173944	0.0201802	1.3615
12	153.238	6949	16495	1	0.9659	0.0179882	0.0236435	2.6103
13	215.037	8340	19978	1	0.9972	0.0227818	0.0194214	2.7902
14	146.853	6827	19239	1	1	0.0155266	0.0215708	2.0429

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	198	514	258.309	737.512	7	5	9
2	55	55	66.1838	236.995	2	1	0
3	224	373	300.238	767.541	7	6	8
4	416	877	530.345	1180.76	10	16	16
5	295	448	364.262	573.215	6	9	9
6	150	332	196.267	606.973	4	6	6
7	69	130	173.468	602.973	2	4	6
8	480	694	566.801	924.879	11	15	9
9	211	456	305.764	751.009	6	9	11
10	314	491	424.363	777.612	8	8	13
11	97	179	169.196	489.073	4	6	6
12	96	197	147.853	448.86	3	5	6
13	182	297	214.924	433.363	6	4	6
14	103	215	125.782	482.801	4	3	3


I534_z25c1-2: distributions, pre-validation, px
Lacuna area (n = 14)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 22343)
count
width (px)